

INSTITUTE OF ACTUARIES OF INDIA

Subject CS2A – Risk Modelling and Survival Analysis (Paper A)

November 2025 Examination

INDICATIVE SOLUTION

Introduction:

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiners have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1: Correct Answer is Option C.

We are calculating the probability of being health after 5 weeks. So here opting out due to health reasons represents D_j and being voted out is a case of censoring C_j .

t	N_j	D_j	C_j	D_j/N_j	$1 - D_j/N_j$
0	10	0	1	0	1
1	9	0	1	0	1
2	8	0	1	0	1
3	7	0	1	0	1
4	6	0	1	0	1

Probability of being healthy after 5 weeks using the Kaplan Meier Estimator

$$\begin{aligned}
 &= p_{KM}(5) \\
 &= 1 * 1 * 1 * 1 * 1 \\
 &= 1
 \end{aligned}$$

Probability of being healthy after 5 weeks using Nelson-Aalan Estimator

$$\begin{aligned}
 &= p_{NA}(5) \\
 &= \exp \left(-\sum \frac{D_j}{N_j} \right) \\
 &= \exp (0) \\
 &= 1
 \end{aligned}$$

So, both probabilities are equal to 1.

[2]

Solution 2: Correct Answer is Option A.

- The straight diagonal line – Model 3 – represents a random classifier as the TPR is equal to FPR
- The point (0,1) – Model 1 – represents a perfect classifier because it has a FPR equal to 0 and TPR equal to 1.
- The curve in the top left – Model 2 – represents a good classifier because it has higher value of TPR against the corresponding value of FPR.
- The curve in the bottom right – Model 4 – represents a bad classifier because it has a lower value of TPR against the corresponding value of FPR.

So, Option A represents correctly matched pairs.

[2]

Solution 3: Correct Answer is Option E.

For calculating central exposed to risk,

- Start Date = Latest of the following:
 - Date of 55th birthday,
 - 1st April 2024
- End Date = Earliest of the following:
 - Date of 56th birthday,
 - Date of Death,
 - 31st March 2025.

Life	Date of 55 th birthday	Date of Death	Start Date	End Date	No of Months
B	1 st July 2023	1 st December 2024	1 st April 2024	30 th June 2024	3

D	1 st December 2023	-	1 st April 2024	30 th November 2024	8
E	1 st February 2024	1 st May 2024	1 st April 2024	1 st May 2024	1
H	1 st August 2024	1 st March 2025	1 st August 2024	1 st March 2025	7
J	1 st January 2025	-	1 st January 2025	31 st March 2025	3
					22

[2]

Solution 4: Correct Answer is Option B.

As the multi-variate time series process is stationary, both the eigen values have to be lower than 1 (in terms of absolute value).

So, the correct eigen values are: $\lambda = 0.57$ and $\lambda = -0.07$

[2]

Solution 5: Correct Answer is Option D.

It is given that k_t decreases linearly from 1.5 at time 0 to -1.5 at time 40 such that from period 0 to 40, $\sum k_t = 0$.

So, at time 20, k_t is expected to take the value 0.

$$\begin{aligned}
 m_{60,20} &= \exp(a_{60} + b_{60} * k_{20}) \\
 &= \exp(a_{60} + 0) \\
 &= \exp(-4.70) \\
 &= 0.00910.
 \end{aligned}$$

[2]

Solution 6: Correct Answer is Option B.

$$\begin{aligned}
 u &= F_X(X=2) = 1 - \exp(-2*0.2) = 1 - \exp(-0.4) = 0.32968 \\
 v &= F_Y(Y=200) = 0.857
 \end{aligned}$$

$$\begin{aligned}
 C(u, v) &= \exp(-[(-\ln u)^2 + (-\ln v)^2]^{1/2}) \\
 &= \exp(-[(-\ln 0.32968)^2 + (-\ln 0.857)^2]^{1/2}) \\
 &= \exp(-1.255099^{1/2}) \\
 &= \exp(-1.120312) \\
 &= 0.326178.
 \end{aligned}$$

[2]

Solution 7: Correct Answer is Option E.

Contribution to partial likelihood for those friends who have resigned

$$\begin{aligned}
 &= \frac{\exp(0+0) * \exp(1+0)}{(\exp(1+1) + \exp(0+0) + \exp(1+0) + \exp(0+1))^2} \\
 &= \frac{1 * \exp(1)}{(1 + \exp(2) + \exp(1) + \exp(1))^2} \\
 &= \frac{\exp(1)}{(1 + \exp(2) + 2\exp(1))^2}
 \end{aligned}$$

[2]

Solution 8: Correct Answer is Option C.

Let median be denoted by m .

$$P(X \leq m) = 0.50$$

$$1 - \exp(-0.0004m^2) = 0.50$$

$$\exp(-0.0004m^2) = 0.50$$

$$-0.0004m^2 = \ln 0.50$$

$$m = 42.$$

[2]

Solution 9: Correct Answer is Option D.

Consider each of the statements:

- | | |
|-----------------------------------|---|
| 1. $E(T_x) = e_x$ | This statement is incorrect as $E(T_x) = e_x^0$ |
| 2. $E(K_x) = e_x^0$ | This statement is incorrect as $E(K_x) = e_x$ |
| 3. $E(T_x) \approx E(K_x) - 0.50$ | This statement is incorrect as $E(T_x) \approx E(K_x) + 0.50$ |
| 4. $E(T_x) \approx E(K_x) + 0.50$ | This statement is correct |
| 5. $T_x \leq K_x$ | This statement is incorrect as $K_x \leq T_x$ |

So, Statement 4 is the only correct statement.

[2]

Solution 10: Correct Answer is Option A.

Required probability

$$= \exp\left(-\int_0^{10} 0.01 t \, dt\right)$$

$$= \exp(-0.01 * \frac{1}{2} * (10^2 - 0^2))$$

$$= \exp(-0.5)$$

$$= 0.6065 = 0.61.$$

[2]

Solution 11: Correct Answer is Option E.

Central Exposed to Risk

$$= \frac{1}{4} * [(10000+9500)/2 + (9500+12500)/2 + (12500+7500)/2 + (7500+10000)/2]$$

$$= \frac{1}{4} * [9750 + 11000 + 10000 + 8750]$$

$$= 9875.$$

Rate of theft for FY 2024-25

$$= 300/9875$$

$$= 3.04\%$$

[2]

Solution 12: Correct Answer is Option D.

The ACF cuts off (becomes 0) at all lags greater than 1, whereas the PACF decays towards 0. Hence X_t is MA (1) process. It is a case of correlated noise.

So, the right answer is ARIMA (0, 0, 1).

[2]

Solution 13: Correct Answer is Option A.

$$\text{Hazard Rate} = (1 - f(x)) / F(x)$$

$$F(x) = 1 - 1/(1+x)$$

$$f(x) = (1 * x^{1-1}) / (1 + x^1)^2 = 1 / (1+x)^2$$

Hazard rate

$$\begin{aligned}
 &= (1 - 1 / (1+x)^2) / (1 - 1 / (1+x)) \\
 &= ((1+x)^2 - 1) / (1+x-1) * (1+x) / (1+x)^2 \\
 &= (1 + 2x + x^2 - 1) / (x (1+x)) \\
 &= (x (2+x)) / (x (1+x)) \\
 &= (2+x) / (1+x).
 \end{aligned}$$

[2]

Solution 14: Correct Answer is Option C.

Just before 12th month, Kaplan-Meier estimate = 0.80

$$\begin{aligned}
 &\text{Revised Kaplan Meier Estimate at the end of 12th month} \\
 &= 0.80 * 1 - (3 / (7+3)) \\
 &= 0.56.
 \end{aligned}$$

[2]

Solution 15: Correct Answer is Option B.

Let us consider each of the given options:

- A. Gumbel Copula has zero lower tail dependence but high upper tail dependence.
- B. Clayton Copula shows high lower tail dependence but zero upper tail dependence.
- C. Gaussian Copula has zero dependence in both the tails.
- D. Frank Copula has zero dependence in both the tails.
- E. Product Copula has zero dependence in both the tails.

So, Clayton Copula would be natural choice for modelling the joint behaviour of oil and natural gas prices as it shows strong lower tail dependence.

[2]

Solution 16: i) $E(N) = \lambda = 100$

$$E(X_i) = (0+200)/2 = 100$$

Let S denote the aggregate annual claims for the cyber insurance portfolio.

$$E(S) = E(N) * E(X_i) = 100 * 100 = 10,000.$$

Savings in aggregate annual claims for the direct insurer is equivalent to the claims ceded in the favour of the reinsurer.

Let S_{QS} denote the reduction in the aggregate annual claims for the cyber insurance portfolio after entering into the quota share arrangement.

$$E(S_{QS}) = 20\% * 10,000 = 2,000.$$

(2)

ii) Under this reinsurance arrangement with Math Re

- Premium per individual risk event would be equal to $100 + 25\% = 125$
- A claim can be triggered with the reinsurer if the individual claim size exceeds $80\% * 125$ i.e. 100.
- Maximum amount that the reinsurer would be bear is up to the coverage limit of $100\% * 125$ i.e. 125.

We need to define N_R which represents the number of payments made by the reinsurer (Math Re).

$$N_R = I_1 + I_2 + I_3 + \dots + I_N$$

N_R is a sum of 'N' indicator (Bernoulli) variables I_j with $j = 1$ to N .

I_j is a Bernoulli (indicator) variable which takes the values –

- 1 with a probability of p where $p = P(X > 100) = (200-100)/200 = 0.50$
- 0 with a probability of $(1 - p) = 1 - 0.50 = 0.50$

I_j has the MGF

$$M_{I_j}(t) = p * \exp(t) + 1 - p$$

$$\begin{aligned} M_{N_R}(t) &= M_N(\log M_{I_j}(t)) \\ &= \exp(\lambda (M_{I_j}(t) - 1)) \\ &= \exp(\lambda (0.50 * \exp(t) + 0.50 - 1)) \\ &= \exp(0.50 \lambda (\exp(t) - 1)) \end{aligned}$$

By uniqueness property of MGFs, it follows that N_R has a Poisson distribution with a parameter equal to $0.50\lambda = 0.50 * 100 = 50$.

It also follows that $E(N_R) = 0.50\lambda = 50$.

So the expected number of claims to be ceded in the favour of the reinsurer is equal to 50. (4)

iii) Let us define Z which represents the individual claim amounts which would be paid by the reinsurer.

Z will take the following values:

Value of Z	Condition
$Z = 0$	if $X < 100$
$Z = X - 100$	if $100 \leq X < 125$
$Z = 25$	if $X > 125$

$$\begin{aligned} E(Z) &= 0 + \int_{100}^{125} (x - 100) / 200 dx + \int_{125}^{200} 25 / 200 dx \\ &= 1/400 * ((25)^2 - (0)^2) + 25/200 * (200 - 125) \\ &= 1.5625 + 9.375 \\ &= 10.9375 \end{aligned}$$

Let S_{SL} denote the reduction in the aggregate annual claims for the cyber insurance portfolio after entering into the quota share arrangement.

$$\begin{aligned} E(S_{SL}) &= E(N) * E(Z) \\ &= 100 * 10.9375 \\ &= 1093.75. \end{aligned}$$

$$E(S_{QS}) > E(S_{SL}).$$

Hence, the quota share reinsurance arrangement with Stat Re leads to higher savings in expected aggregated annual claims as compared to the stop loss reinsurance arrangement with Math Re. (3)

iv) Let us consider the all-possible situations:

	Loss Ratio (LR)	Claims as a % of premiums	
		First Arrangement Attachment Ratio 70%, Coverage Limit 90%	Second Arrangement Attachment Ratio 60%, Coverage Limit 80%
1	$LR \leq 60\%$	= 0%	= 0%
2	$60\% < LR \leq 70\%$	= 0%	$\leq 10\%$
3	$70\% < LR \leq 80\%$	$\leq 10\%$	$\leq 20\%$
4	$80\% < LR \leq 90\%$	$\leq 20\%$	= 20%
5	$LR > 90\%$	= 20%	= 20%

The statement made by the intern is true only in the last situation. In all other situations, the claims as a percentage of premiums passed on to the reinsurer are different across both the stop loss arrangements. So, both the arrangements are not equivalent.

(1)
[10]

Solution 17: i)

$$\begin{aligned}
 R_t &= \log(P_t / P_{t-1}) \\
 &= \log(P_{t-1} * \exp(\mu + e_t) / P_{t-1}) \\
 &= \log(\exp(\mu + e_t)) \\
 &= \mu + e_t
 \end{aligned}$$

Thus, R_t is a moving average MA (0) process.

(1)

ii)

$$\begin{aligned}
 E(R_t) &= E(\mu + e_t) \\
 &= \mu.
 \end{aligned}$$

Thus, $E(R_t)$ is constant and independent of t . ----- (1)

$$\begin{aligned}
 \text{Var}(R_t) &= \text{Cov}(R_t, R_t) \\
 &= \text{Cov}(\mu + e_t, \mu + e_t) \\
 &= \sigma^2
 \end{aligned}$$

Thus, $\text{Var}(R_t)$ is a constant variance. ----- (2)

$$\begin{aligned}
 \text{Cov}(R_t, R_{t-k}) &= \text{Cov}(\mu + e_t, \mu + e_{t-k}) \\
 &= 0 \dots\dots\dots \text{as } e_t \text{ and } e_{t-k} \text{ are independent of each other and hence covariance is 0.} \\
 \text{Thus, } \text{Cov}(R_t, R_{t-k}) &\text{ is also constant and it does not depend on time } t \text{ but only on the lag } k. \text{ ----- (3)}
 \end{aligned}$$

Considering results (1), (2) and (3), we can conclude that R_t is a (weakly) stationary time series process.

(3)

iii)

$$\begin{aligned}
 \text{Var}(X_t) &= \text{Var}(b * X_{t-1} + e_t) \\
 &= b^2 * \text{Var}(X_{t-1}) + \text{Var}(e_t) \dots\dots\dots (1)
 \end{aligned}$$

As $e_t \sim N(0, \sigma_t^2)$,

$$\begin{aligned} \text{Var}(e_t) &= E(\sigma_t^2) \quad \dots\dots\dots \text{the variance of the error terms is the expected value of } \sigma_t^2 \\ &= E(a_0 + a_1 * e_{t-1}^2) \\ &= a_0 + a_1 * E(e_{t-1}^2) \quad \dots\dots\dots (2) \end{aligned}$$

$$\begin{aligned} \text{Var}(e_{t-1}) &= E(e_{t-1}^2) - E^2(e_{t-1}) \\ &= E(e_{t-1}^2) - 0 \quad \dots\dots\dots \text{As } e_t \sim N(0, \sigma_t^2), E^2(e_{t-1}) = 0 \\ &= E(e_{t-1}^2) \quad \dots\dots\dots (3) \end{aligned}$$

From (2) and (3),

$$\begin{aligned} \text{Var}(e_t) &= a_0 + a_1 * E(e_{t-1}^2) \\ &= a_0 + a_1 * \text{Var}(e_{t-1}) \end{aligned}$$

Let $\text{Var}(e_t) = \text{Var}(e_{t-1}) = v$

$$\begin{aligned} v &= a_0 + a_1 * v \\ v &= a_0 / (1 - a_1) \quad \dots\dots\dots (4) \end{aligned}$$

From (1) and (4),

$$\begin{aligned} \text{Var}(X_t) &= b^2 * \text{Var}(X_{t-1}) + \text{Var}(e_t) \\ &= b^2 * \text{Var}(X_{t-1}) + a_0 / (1 - a_1) \end{aligned}$$

Let $\text{Var}(X_t) = \text{Var}(X_{t-1}) = V_x$

$$V_x = b^2 * V_x + a_0 / (1 - a_1)$$

$$\begin{aligned} V_x &= a_0 / ((1 - a_1)(1 - b^2)) \\ \text{Var}(X_t) &= a_0 / ((1 - a_1)(1 - b^2)) \quad \dots\dots\dots \text{hence, proved.} \end{aligned} \quad (4)$$

iv)

$$\begin{aligned} \text{ACF at lag 1} &= \text{Cov}(X_t, X_{t-1}) / \text{Var}(X_t) \quad \dots\dots\dots (1) \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_t, X_{t-1}) &= \text{Cov}(b * X_{t-1} + e_t, X_{t-1}) \\ &= b * \text{Var}(X_{t-1}) + 0 \quad \dots\dots\dots \text{As } e_t \text{ is independent of } X_{t-1} \\ &= b * \text{Var}(X_{t-1}) \quad \dots\dots\dots (2) \end{aligned}$$

From (1) and (2),

$$\begin{aligned} \text{ACF at lag 1} &= b * \text{Var}(X_{t-1}) / \text{Var}(X_t) \\ &= b \quad \dots\dots\dots \text{As } \text{Var}(X_{t-1}) = \text{Var}(X_t) \end{aligned}$$

Hence, the autocorrelation function at lag 1 for the new time series process X_t is equal to b .

(2)
[10]

Solution 18: i) Let the number of exotic birds the music composer is able to watch during a week be X .

$$X \sim \text{Bin}(5, 0.50)$$

$$P(X=0) = {}^5C_0 * 0.50^0 * 0.50^5 = 0.03125$$

$$P(X=1) = {}^5C_1 * 0.50^1 * 0.50^4 = 0.15625$$

$$P(X \geq 2) = {}^5C_2 * 0.50^2 * 0.50^3 = 0.81250$$

Probability of composing no song during the week = $P(X=0) = 0.03125$

Probability of composing one song during the week = $P(X=1) = 0.15625$

Probability of composing two songs during the week = $P(X \geq 2) = 0.8125$

Probability of composing three or more songs in a week = 0

(3)

ii) No of songs at the end of a particular week

= No of songs at the beginning of the week + No of new songs received from film producer – Number of songs (to be) composed during the week – Number of songs returned to the film producer (due to inability to compose them).

	No of songs at end of prev. week	No of songs at end of next week	No of songs received during the week	No of songs to be composed in this week	No of songs returned to the producer	Probability
p_{00}	0	0	2	2	0	$P(X \geq 2) = 0.8125$
p_{01}	0	1	2	1	0	$P(X=1) = 0.15625$
p_{02}	0	2	2	0	0	$P(X=0) = 0.03125$
p_{10}	1	0	2	>2	NA #	0
p_{11}	1	1	2	2	0	$P(X \geq 2) = 0.8125$
p_{12}	1	2	2	1	0	$P(X=1) = 0.15625$
	1	2	2	0	1	$P(X=0) = 0.03125$
p_{20}	2	0	2	>2	NA #	0
p_{21}	2	1	2	>2	NA #	0
p_{22}	2	2	2	2	0	$P(X \geq 2) = 0.8125$
	2	2	2	1	1	$P(X=1) = 0.15625$
	2	2	2	0	2	$P(X=0) = 0.03125$

The excess songs can be returned to the producer only if the number of pending songs exceeds two and after returning these songs, the number of pending songs is brought down to two songs. So, in case of transitions from state 1 to state 0, state 2 to state 0 or state 2 to state 1, return of songs is not possible because the end state (no of pending songs after returning the excess) is lower than 2.

Errors:

1. **Error: $P(0,0) = 0.31250$**

$P(0,0)$ is based on the probability of composing 2 songs during the week which corresponds to probability of watching 2 or more exotic birds during the week.

Wrong Value: $P(X=2) = 0.31250$ has been considered here – which is the probability of watching exactly two birds during the week.
Also, the sum of transition probabilities from state 0 is less than 1.

Rectified Value: $P(X \geq 2) = 0.81250$ should be considered here – which is the probability of watching 2 or more birds during the week.
After rectification of error 1, the sum of transition probabilities from state 0 would be equal to 1.

2. Error: $P(1,0) = 0.31250$

$P(1,0)$ represents a situation wherein there was 1 song pending at the beginning and there would no songs pending at the end. It is not possible to return the songs to the producer in this case as the end state is lower than 2. Hence, this would mean expecting the composer to compose 3 songs during the week which is not possible.

Wrong Value: $P(X=3) = 0.31250$ has been considered here – which is the probability of watching exactly three birds during the week.
Also, the total of transition probabilities from state 1 exceeds one.

Rectified Value: 0 as this particular transition from state 1 to state 0 is not possible in the given circumstances.
After rectification of error 2, the sum of transition probabilities from state 1 would be equal to 1.

3. Error: $P(2,0) = 0.15625$

$P(2,0)$ represents a situation wherein there were 2 songs pending at the beginning and there would no songs pending at the end. It is not possible to return the songs to the producer in this case as the end state is lower than 2. Hence, this would mean expecting the composer to compose 4 songs during the week which is not possible.

Wrong Value: $P(X=4) = 0.15625$ has been considered here – which is the probability of watching exactly four birds during the week.
Also, the total of transition probabilities from state 2 exceeds one.

Rectified Value: 0 as this particular transition from state 2 to state 0 is not possible in the given circumstances.
After rectification of errors 3 and 4, the sum of transition probabilities from state 2 would be equal to 1.

4. Error: $P(2,1) = 0.31250$

$P(2,1)$ represents a situation wherein there were 2 songs pending at the beginning and there would 1 song pending at the end. It is not possible to return the songs to the producer in this case as the end state is lower than 2. Hence, this would mean expecting the composer to compose 3 songs during the week which is not possible.

Wrong Value: $P(X=3) = 0.31250$ has been considered here – which is the probability of watching exactly three birds during the week.

Also, the total of transition probabilities from state 2 exceeds one.

Rectified Value: 0 as this particular transition from state 2 to state 1 is not possible in the given circumstances.

After rectification of errors 3 and 4, the sum of transition probabilities from state 2 would be equal to 1.

The rectified transition matrix is given below:

		No of songs pending to be composed at the end of the next week		
		0	1	2
No of songs pending to be composed at the end of the previous week	0	0.8125	0.15625	0.03125
	1	0	0.8125	0.1875
	2	0	0	1

The detailed calculation of probabilities and the rectified transition matrix is not required to be a part of the answer but given here for completeness of the solution.

(5)

iii) Required Probability

$$= P(0,1) * P(1,2) + P(0,0) * P(0,2) + P(0,2) * P(2,2)$$

$$= 0.15625 * 0.1875 + 0.8125 * 0.03125 + 0.03125 * 1$$

$$= 0.085938.$$

(2)

[10]

Solution 19: i) Graph under the first method – graduation using parametric formula is not a good fit to the data as the graduated default rates are much lower than the actual default rates for majority of the durations.

Graph under the second method – graduation using spline functions is actually optimal with much lower differences between actual and expected default rates.

So, second method is a better fit to the underlying data.

(1)

ii) Formula under *Method 1: Gompertz Law*:

$$\mu_x = B * c^x$$

Formula under *Method 2: Spline-based Graduation with three knots*:

$$\mu_x = \alpha_0 + \alpha_1 * x + \beta_1 * \Phi_1(x).$$

(2)

iii)

Null and Alternative Hypotheses for both the methods:

H₀: the graduated rates are the true underlying default rates for the corporate bonds.

H₁: the graduated rates are not the true underlying default rates for the corporate bonds.

Degrees of freedom under both the methods:

Method 1:

In this method using parametric formula, we are estimating two parameters. Hence, the degrees of freedom for testing the fit of the first method would be equal to $15 - 2 = 13$.

Method 2:

In this method using cubic spline functions, we are using three knots and hence estimating three parameters. Hence, the degrees of freedom for testing the fit of the second method would be equal to $15 - 3 = 12$.

Critical value under both the methods:*Method 1:*

Critical value at 13 degrees of freedom at 0.5% level of significance = 29.819

Method 2:

Critical value at 12 degrees of freedom at 0.5% level of significance = 28.299

Conclusion for Method 1:

In respect of method 1, as the value of the test statistic (86.34) is much greater than the critical value (29.819), we have sufficient evidence to reject the null hypothesis at 0.5% level of significance. Hence, the graduation using parametric formula is not a good fit for the data. This is in line with the graph from which it is visually evident that method 1 is tending to underestimate default rates at majority of the durations.

Conclusion for Method 2:

In respect of method 2, as the value of the test statistic (33.09) is marginally greater than the critical value (28.299), we have sufficient evidence to reject the null hypothesis at 0.5% level of significance. Hence, the graduation using spline functions with three knots is also not a good fit for the data. However, it is much better than method 1 (parametric formula) as the difference between test statistic value and critical value is much less.

Even in case of 10 degrees of freedom, as $33.09 > 25.19$, we have sufficient evidence to reject the null hypothesis at 0.5% level of significance.

(5)

iv)

- First defect is addressed by “Standardized Deviations Test”.
- Second defect is addressed by “Signs Test”.

(2)

[10]

Solution 20:

i) Following is the required one-step transition matrix:

	L1	L2	L3	L4	L5	L6
L1	0	A=1/2	B=1/2	0	0	0
L2	C=1/6	D=2/3	0	E=1/6	0	0
L3	0	0	F=2/3	G=1/3	0	0
L4	0	0	H=2/3	I=1/6	0	J=1/6
L5	0	2/3	0	1/6	0	1/6
L6	0	0	0	0	0	1

Explanation:

- From L1:

- Either moves to L2 by rolling an even number (probability = $\frac{1}{2}$) or moves to L3 by rolling an odd number (probability = $\frac{1}{2}$).
- All other transition probabilities in the row are equal to 0.
- **From L2:**
 - By rolling a prime number, player moves to the prime number displayed minus one. Otherwise, player stays at L3.
 - Prime numbers are 2, 3 and 5. So player can move either to L1, L2 and L4 with probability of $\frac{1}{6}$.
 - Player stays in L2 otherwise with a probability of $\frac{1}{2}$.
 - Total probability of staying in L2 = $\frac{2}{3}$ ($\frac{1}{2} + \frac{1}{6}$).
 - All other transition probabilities in the row are equal to 0.
- **From L3:**
 - By rolling a perfect square, player moves to L4, otherwise player stays at L3.
 - Perfect squares are 1 and 4. So player can move to L4 with a probability of $\frac{1}{3}$ ($\frac{1}{6} + \frac{1}{6}$).
 - Player stays at L3 with a probability of $\frac{2}{3}$.
 - All other transition probabilities in the row are equal to 0.
- **From L4:**
 - By rolling a composite number, player moves to the composite number being displayed. Otherwise, player moves to L3.
 - Composite numbers are 4 and 6.
 - Player can move to L6 with a probability of $\frac{1}{6}$.
 - Player can stay at L4 with a probability of $\frac{1}{6}$.
 - Player moves to L3 otherwise with a probability of $\frac{2}{3}$.
 - All other transition probabilities in the row are equal to 0.
- **From L5:**
 - By rolling a composite number, player moves to the composite number being displayed. Otherwise, player moves to L2.
 - Composite numbers are 4 and 6.
 - Player can move to L6 with a probability of $\frac{1}{6}$.
 - Player can move to L4 with a probability of $\frac{1}{6}$.
 - Player moves to L2 otherwise with a probability of $\frac{2}{3}$.
 - All other transition probabilities in the row are equal to 0.
- **From L6:**
 - Once at L6, the game is over. So, player stays in L6 with probability of 1.
 - All other transition probabilities in the row are equal to 0.

(5)

ii) Let us define N_i to represent the expected number of throws of die required to reach the final treasure from the current state 'i' where $i = L1, L2, L3, L4, L5, L6$.

We get the following equations:

1. $N_{L6} = 0$
2. $N_{L1} = 1 + \frac{1}{2} * N_{L2} + \frac{1}{2} * N_{L3}$
3. $N_{L2} = 1 + \frac{1}{6} * N_{L1} + \frac{1}{6} * N_{L4} + \frac{2}{3} * N_{L2}$
 $N_{L2} = \frac{1}{2} * N_{L1} + \frac{1}{2} * N_{L4} + 3$
4. $N_{L3} = 1 + \frac{1}{3} * N_{L4} + \frac{2}{3} * N_{L3}$

$$N_{L3} = N_{L4} + 3$$

$$\begin{aligned} 5. \quad N_{L4} &= 1 + 1/6 * N_{L4} + 1/6 * N_{L6} + 2/3 * N_{L3} \\ N_{L4} &= 6/5 + 1/5 * N_{L6} + 4/5 * N_{L3} \\ N_{L4} &= 6/5 + 4/5 * N_{L3} \quad \text{..... From (1) } N_{L6} = 0 \end{aligned}$$

Solving equations (4) and (5) simultaneously,

$$\begin{aligned} 6. \quad N_{L4} &= 6/5 + 4/5 * (N_{L4} + 3) \\ N_{L4} &= 18 \end{aligned}$$

From (4) and (7),

$$7. \quad N_{L3} = N_{L4} + 3 = 18 + 3 = 21$$

Substitute the value of N_{L3} and N_{L4} into equations (2) and (3),

$$8. \quad N_{L1} = 1 + 1/2 * N_{L2} + 21/2$$

$$\begin{aligned} 9. \quad N_{L2} &= 1/2 * N_{L1} + 9 + 3 \\ N_{L2} &= 1/2 * N_{L1} + 12 \end{aligned}$$

Solving equations (8) and (9) simultaneously

$$\begin{aligned} 10. \quad N_{L1} &= 1 + 1/2 * (1/2 * N_{L1} + 12) + 21/2 \\ 2N_{L1} &= 2 + 1/2 N_{L1} + 12 + 21 \\ 3/2 N_{L1} &= 35 \\ N_{L1} &= 70/3 = 23.33 \end{aligned}$$

Thus, the expected number of throws of die required to reach to the final treasure is 23.33.

(5)
[10]

Solution 21: i) Let us define the values that each covariate can take:

$Z_1 = 1$ for	Superconducting qubits	$Z_1 = 0$ for	Trapped ion qubits
$Z_2 = 1$ for	Yes (Error Correction)	$Z_2 = 0$ for	No (Error Correction)
$Z_3 = 1$ for	Cryogenic cooling	$Z_3 = 0$ for	Dilution fridge method

Let us now define the values of regression parameters for each variate:

$$\begin{aligned} \beta_1 &= (0.45+0.55)/2 = 0.50 \\ \beta_2 &= (-0.80-0.60)/2 = -0.70 \\ \beta_3 &= (-0.50-0.10)/2 = -0.30 \end{aligned} \quad (2)$$

ii) Class of qubits to which the baseline hazard applies to:

- trapped ion qubits,
 - with no error correction and
 - which use dilution fridge cooling method
- (1)

iii)

- Superconducting qubit type has a positive value of regression parameter. Hence it contributes to the hazard and thus leads to faster decoherence as compared to trapped ion qubits. This statement is TRUE.

- Error correction has a negative value of regression parameter. Hence it reduces the hazard and thus contributes to survival of the qubits. This statement is also TRUE.
- Cryogenic cooling has a negative value of regression parameter. Hence it reduces the hazard as compared to dilution fridge method. Hence, this statement is NOT TRUE / FALSE.

(3)

iv) Probability of survival for a trapped ion qubit which was subject to error correction and cryogenic cooling in 4 milliseconds

$$\begin{aligned}
 &= \exp \left(- \int_0^4 H_0(t) * \exp (\beta_1 * Z_1 + \beta_2 * Z_2 + \beta_3 * Z_3) dt \right) \\
 &= \exp \left(- \int_0^4 H_0(t) * \exp (0 - 0.70 * 1 - 0.30 * 1) dt \right) \\
 &= \exp \left(- \int_0^4 H_0(t) * \exp (-1) dt \right) \dots\dots\dots \mathbf{A} = -1 \\
 &= (\exp \left(- \int_0^4 H_0(t) dt \right) \exp (-1)) \dots\dots\dots \mathbf{B} = \exp(-1)
 \end{aligned}$$

$$\text{Let } \exp \left(- \int_0^4 H_0(t) dt \right) = x$$

$$x^{\exp (-1)} = 0.60$$

Probability of survival for a superconducting type qubit which was not subject to error correction but subject to dilution fridge cooling in 4 milliseconds

$$\begin{aligned}
 &= \exp \left(- \int_0^4 H_0(t) * \exp (\beta_1 * Z_1 + \beta_2 * Z_2 + \beta_3 * Z_3) dt \right) \\
 &= \exp \left(- \int_0^4 H_0(t) * \exp (0.50 * 1 + 0 + 0) dt \right) \\
 &= \exp \left(- \int_0^4 H_0(t) * \exp (0.50) dt \right) \dots\dots\dots \mathbf{C} = 0.50 \\
 &= \exp \left(- \int_0^4 H_0(t) dt \right) \exp (0.50) \\
 &= x^{\exp (0.50)} \\
 &= x^{\exp (-1+1.50)} \\
 &= x^{\exp (-1) \exp (1.50)} \\
 &= (x^{\exp (-1)})^{\exp (1.50)} \\
 &= 0.60^{\exp (1.50)} \dots\dots\dots \mathbf{D} = 0.60 \\
 &= 0.101331
 \end{aligned}$$

(4)

[10]

Solution 22: i) From going from State T to State P, it would take 2 years.
If one goes directly from State T to State F, it would also take 2 years.

So, the average waiting time in State T is 2 years.

Once the process is in State P, it would take 5 years to go to State F.

So, the average waiting time in State P is 5 years.

Waiting time in States P and T are exponentially distributed with means 2 and 5 respectively.

Waiting Time (State T) ~ Exponential (1/2)
Waiting Time (State P) ~ Exponential (1/5)

(2)

ii) Transition intensity $\mu_{ij} = \lambda_i * p_{ij}$

Where:

- λ_i is the exponential parameter associated with the holding time in state i.

- p_{ij} is the probability of transition from state i to state j .

So, we calculate:

$$\mu_{TP} = \theta = \frac{1}{2} * 0.50 = 0.25$$

$$\mu_{TF} = \beta = \frac{1}{2} * 0.50 = 0.25$$

$$\mu_{PF} = \alpha = \frac{1}{5} * 1 = 0.20. \quad (2)$$

iii) The Kolmogorov's forward differential equation for:

$$\begin{aligned} \frac{d}{dt}(p_{TP}(t)) &= p_{TT}(t) * \mu_{TP} + p_{TP}(t) * \mu_{PP} \\ &= p_{TT}(t) * \mu_{TP} + p_{TP}(t) * (0 - \mu_{PF}) \\ &= \theta * p_{TT}(t) + (0 - \alpha) * p_{TP}(t) \quad \dots \dots \dots \text{Total of each row in generator matrix} = 0 \\ &= 0.25 * p_{TT}(t) - 0.20 * p_{TP}(t). \\ &= 0.25 * e^{-0.5t} - 0.20 * p_{TP}(t). \end{aligned} \quad (2)$$

iv)

- Integrating factor to be used is $\exp(0.2t)$
 - Value of integration constant $c = 5/6$
 - Values of A and B are -0.2 and -0.5 respectively.
- (4)

Detailed calculations have been shown below:

$$\frac{d}{dt}(p_{TP}(t)) = 0.25 * p_{TT}(t) - 0.20 * p_{TP}(t)$$

$$\frac{d}{dt}(p_{TP}(t)) + 0.20 * p_{TP}(t) = 0.25 * p_{TT}(t)$$

Multiplying both sides by the integrating factor $\exp(0.2t)$, we get,

$$\frac{d}{dt}(p_{TP}(t)) * \exp(0.2t) + 0.20 * p_{TP}(t) * \exp(0.2t) = 0.25 * p_{TT}(t) * \exp(0.2t)$$

We use the $p_{TT}(t) = \exp(-0.5t)$,

$$\begin{aligned} \frac{d}{dt}(p_{TP}(t)) * \exp(0.2t) + 0.20 * p_{TP}(t) * \exp(0.2t) &= 0.25 * \exp(-0.5t) * \exp(0.2t) \\ \frac{d}{dt}(p_{TP}(t)) * \exp(0.2t) + 0.20 * p_{TP}(t) * \exp(0.2t) &= 0.25 * \exp(-0.3t) \end{aligned}$$

$$\frac{d}{dt}(p_{TP}(t) * \exp(0.2t)) = 0.25 * \exp(-0.3t)$$

Integrating both sides, we get,

$$p_{TP}(t) * \exp(0.2t) = 0.25/-0.3 * \exp(-0.3t) + c$$

At time $t=0$, by definition, $p_{TP}(t) = 0$.

$$p_{TP}(0) * \exp(0) = 0.25/-0.3 * \exp(0) + c$$

$$0 = -5/6 + c$$

$$c = 5/6$$

$$p_{TP} * \exp(0.2t) = -5/6 * \exp(-0.3t) + 5/6$$

Dividing both sides by $\exp(0.2t)$,

$$p_{TP}(t) = -5/6 * \exp(-0.5t) + 5/6 * \exp(-0.2t)$$

$$p_{TP}(t) = 5/6 * (\exp(-0.2t) - \exp(-0.5t))$$

[10]
